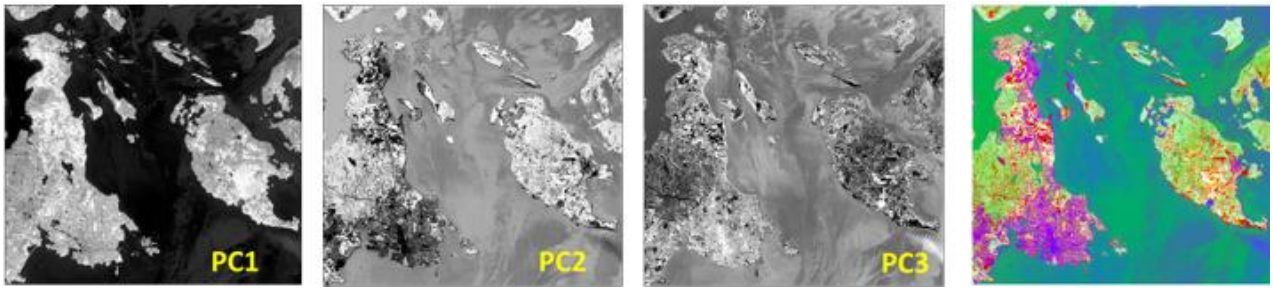
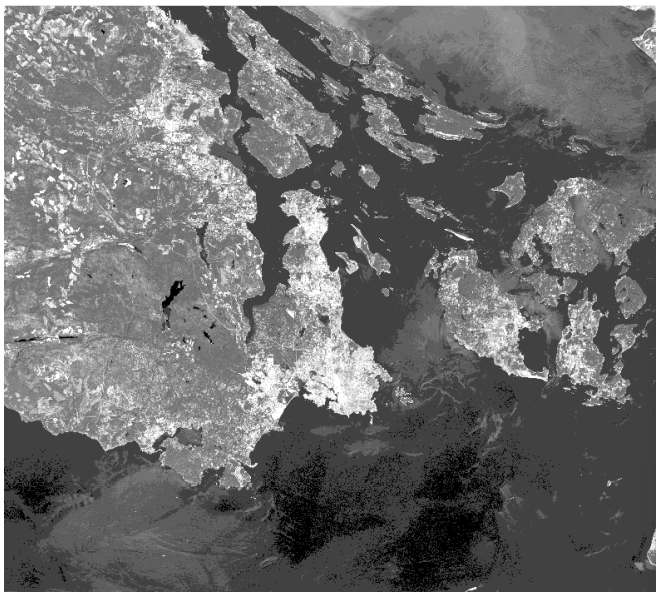



## Geog 228 – Lab 5 Assignment: Enhancements & Transformations

**[25 marks total]**

1. Use the Add Layer Wizard to create a Grayscale of Band 2 from your 8-bit image. Make sure the Tail Trim is set to the default of 2%:
   Explain what happens to the high and low ends of the histogram (hint – use the Edit LUTs tool), and also visually in the display, when you apply the root enhancement. Compare it to what happens when you apply the linear enhancement – *provide screenshots of the grayscale images, AND the histograms showing both enhancements, and refer to specific cover types in the image to help your explanation*. [6 marks]



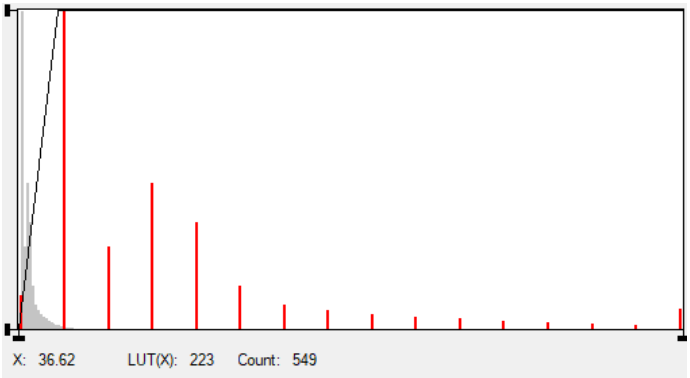



***Figure 1.*** A LANDSAT 9 image of lower Vancouver Island, shown as a greyscale composite for band 4 (red). The image is enhanced linearly with a 2% tail trim on the minimum and maximum outlier values. The histogram compares the raw data digital number values with the enhanced values. Note the histogram is linearly stretched across the entire range of DN values. The greyscale image and histogram were created with CATALYST Professional software.

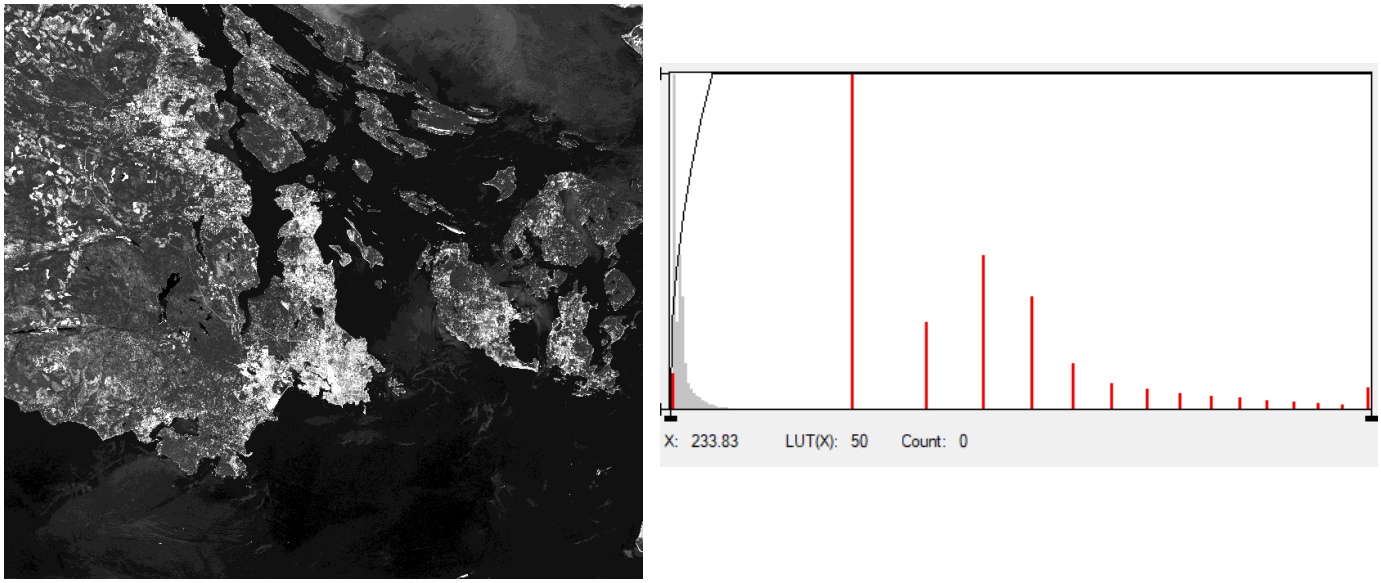



***Figure 2.*** A LANDSAT 9 image of lower Vancouver Island, shown as a greyscale composite for band 4 (red). The image has a root enhancement applied, with a 2% tail trim on the minimum and maximum outlier values. The histogram compares the raw data digital number values with the enhanced values. Note the histogram is stretched across the entire range of DN values. The higher DN values are compressed and the lower values are expanded. This allows for better analysis in the darker areas of the image (Evans, 2023, p.8). The greyscale image and histogram were created with CATALYST Professional software.

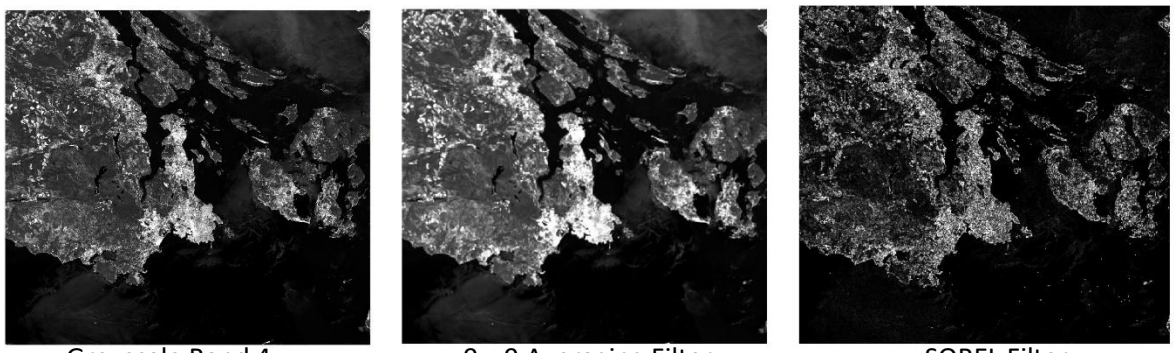
The 2% tail trim removes the top and bottom 2% of values from the raw data and then the enhancement is applied to the remaining 96% of values. The histogram shows the stretching of the raw DN values through the full range of possible values.

The linear enhancement linearly stretches the remaining 96% of the DN values in the image, allowing the whole image to show more grayscale detail. This results in an image where one can more easily analyze the focus are because the overall detail is more pronounced, compared to the raw image (Evans, 2023, p.8). The root enhancement enhances the remaining 96% of values, but instead of a linear stretch, the darker values are compressed and the higher values are expanded. As a result, the image allows for comprehensive analysis in the darker areas and also preserves detail in higher DN range, according to Evans (2023, p.8).

The vegetation in the linearly enhanced image is low contrast and may present difficulties in analysis. The same vegetation land cover is shown is greater detail in the root enhanced image. The urban areas in the linearly enhanced image appear bright and it is difficult to tell what objects may be present in those areas, due to the low contrast. The root image maintains those brighter areas, while allowing the darker areas to be show in more detail. The histograms for both enhancements have a nearly identical distribution of DN values in roughly

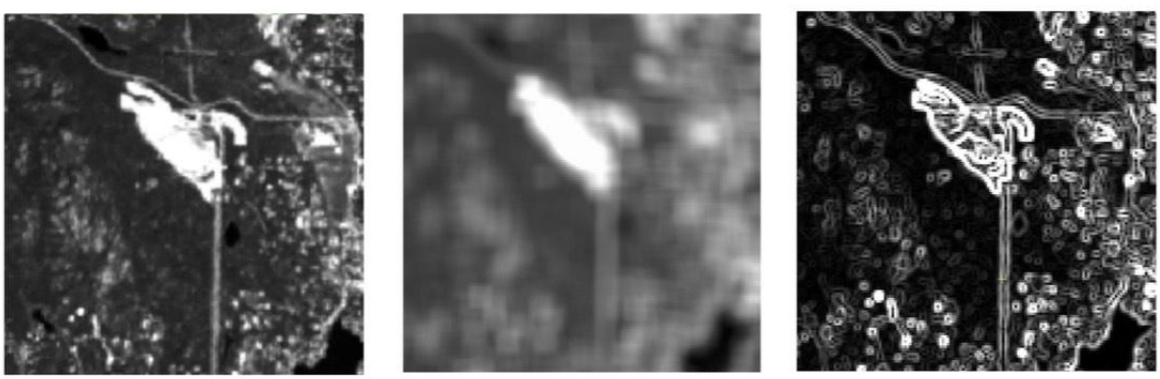
the higher 50% of values, but the lower halves are stretched differently. The ocean water in certain areas of the linearly stretched image resemble the same shade as the vegetation but in the root image, those same areas are easily distinguishable as water.

2. Consider your 2 spatial filter output layers (**SOBEL**, and the **9 x 9** averaging filter – you can exclude the 3 x 3 averaging filter from this analysis):
   Find a **high frequency feature** in the image and explain how the 2 filters have changed the visual appearance of the feature from what you see in the original image, **and WHY**. Create a Figure for your report showing the original image, SOBEL and averaging filter for comparison. **Identify** the high frequency feature with a zoomed in subset and/or drawing a box or circle around it to identify it. [6 marks]



Greyscale Band 4 | 9 x 9 Averaging Filter | SOBEL Filter

***Figure 3.*** A LANDSAT 9 image of lower Vancouver Island in Band 4, depicted three ways. An unfiltered image allows comparison to be done between a source and altered images, to analyze ground cover and feature on the surface. The 9 x 9 averaging filter is used to enhance areas with low contrast changes over distances in the image, known as low frequency features. A SOBEL filter is beneficial for augmenting abrupt changes in adjacent or near pixels, called high frequency features. The image and its filters were created with CATALYST Professional software.



Greyscale Band 4 | 9 x 9 Averaging Filter | SOBEL Filter

***Figure 4.*** A LANDSAT 9 image of the Hartland Landfill on lower Vancouver Island, demonstrating the smoothing and augmentation of high frequency features, through a low and high pass filter, respectively. The image and filters were created with CATALYST Professional software.

A low pass and a high pass filter are used to analyze this image of lower Vancouver Island. An averaging filter is considered a low pass filter and SOBEL is considered a high pass filter. A low pass filter is used to enhance areas with low contrast changes over distances in the image, known as low frequency features. On the contrary, a high pass filter is beneficial for augmenting abrupt changes in adjacent or near pixels, called high frequency features (Evans, 2023, p. 15). A low pass filter may be useful to analyze areas that are homogenous or to reduce image noise. An unfiltered photo may appear somewhat homogenous and a low pass filter may validate the hypothesis that a given area is indeed homogenous. A low pass image will look smooth, homogenized, and blurry. This highlights large areas, rather than small ones with abrupt change in pixel values. A high pass filter is beneficial to emphasize high frequency features, such as coast lines, clear cut areas in forests, fault lines, and urban/developed areas (Evans, 2023, p. 15). A high pass filtered image will appear to have many cell-like shapes, with sharp borders and emphasized lines, in areas where the original image had high frequency filters.

*Figure 4* demonstrates the emphasis of high frequency features by showing the Hartland Landfill with an averaging filter and a SOBEL filter. The pixel DN values with sharp changes are emphasized in the SOBEL image and are smoothed in the averaging filter image.

3. Submit screenshots of your **Simple Ratio** and **NDVI** grayscale layers, as shown in the map display. For both layers, look up and report the **minimum** and **maximum** output pixel values. What are the **advantages** of using an NDVI over a simple ratio? ***(You may have to go outside of the lab manual to answer this question – properly cite your source).*** [7 marks]
   SR – less bright vegetation values
   NDVI – bright vegetation layers

Simple Vegetation Ratio (NIR/Red)

Normalized Difference Vegetation Index (NDVI)

***Figure 5.*** A LANDSAT 9 image of lower Vancouver Island, shown with a simple vegetation ratio and a normalized difference vegetation index. These band ratios/indices aim to minimize any sunlight and shadows that may have occurred during data acquisition and to highlight information. The ratios can also highlight information about vegetation that is not available with only one band (Evans, 2023, p. 18). The images were created with CATALYST Professional software.

The minimum and maximum pixel values for the simple vegetation ratio (SR) and the normalized difference vegetation index (NDVI) respectively are, 0.712426 and 5.33726 versus -0.179269 and 0.684406. These values were sourced from the CATALYST Professional DN value histogram.

The simple vegetation ratio formula is a ratio of the NIR/Red bands and the normalized difference vegetation index formula is (NIR - Red)/(NIR + Red). According to Osbey (2023), the "red reflectance is sensitive to chlorophyll content and the near-infrared reflectance is sensitive to the mesophyll structure of leaves." A large ratio value will represent green vegetation presence, while a low ratio represents the presence pf bare soil or other land cover types that are not green.

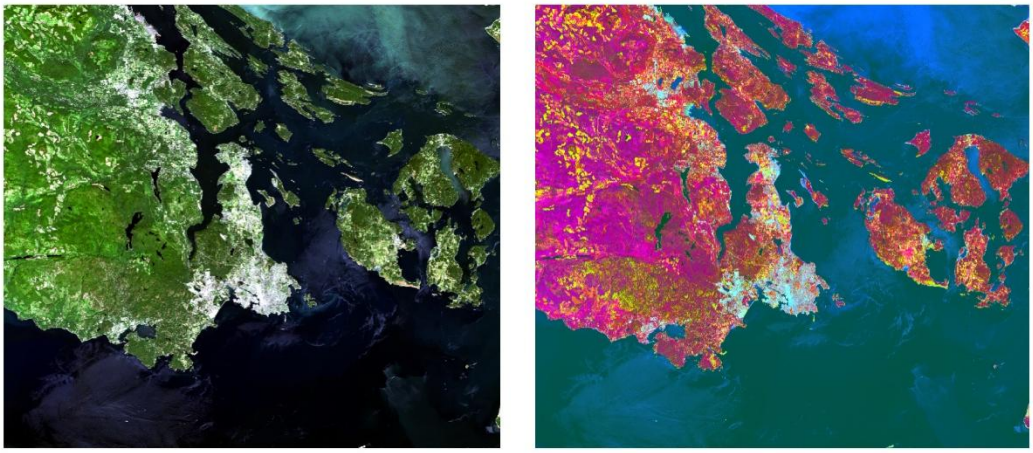
An SV will result in a value of roughly 1, if the NIR and Red Band values are approximately the same. This over simplification can cause complications in land cover classification. As a result, other ratios have been developed to account for various factors.

The NDVI formula is the difference between the NIR and Red values, over their sum. This formula helps account for the irradiance effect. These values always lie between -1 and 1. Values closer to 1 represent areas with large amounts of vegetation, with high NIR reflectance. Values in the middle represent bare soil or rocks (Osbey, 2023). Using the SR values, these areas may have been mistaken for urban areas, depending on context clues in the surrounding areas. Pixels and areas with high amounts of red reflectance, such as water, clouds, and snow will result in values near -1 (Osbey, 2023). The three-tiered nature of NDVI allows for more complex analysis in vegetation land cover classification, as opposed to SR.

*4.* How much of the data variance from the original Landsat image bands is contained <u>eigenchannel 1</u> of your PCA output?

- Submit a screenshot of your RGB composite showing the first 3 Eigen Channels – zoomed to show the full extent of the image.
- Discuss what cover types are easier to distinguish using this PCA output over a true colour composite
- *Provide at least 2 examples illustrating your discussion points (zoomed in subset and/or drawing a box or circle around it – your figure should compare original true colour and the PCA RGB composite).*

[6 marks]

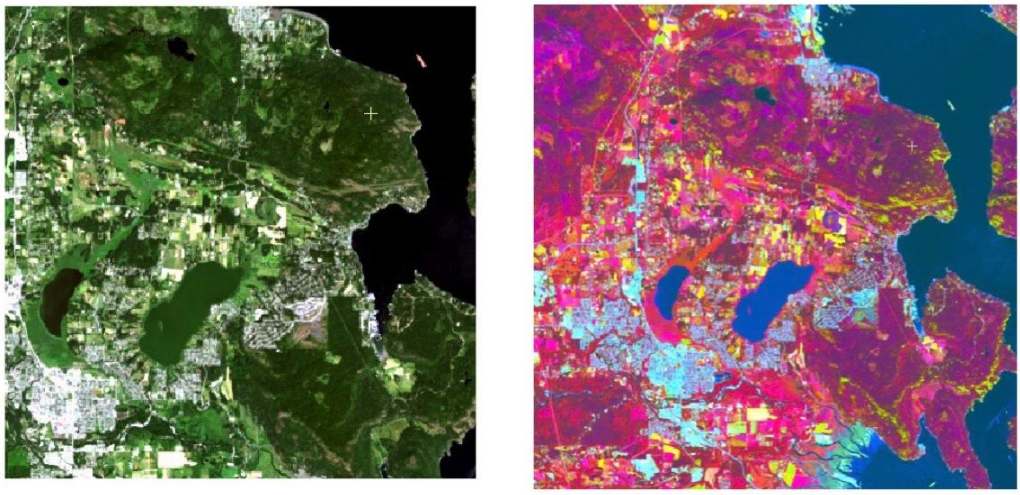


True Colour RGB PCA RGB

***Figure 6.*** A LANDSAT 9 image of lower Vancouver Island, shown as a true colour composite and as a Principal Component Analysis (PCA) RGB composite. The PCA RGB composite is a layered image of three eigen channels, where the channels represent the first three principal components from the original LANDSAT 9 bands. The composites were created with CATALYST Professional software.

```
Run PCA
PCA                                                    2223 EASI/PACE 15:37 14Mar2025

C:\Users\meijsg\AppData\Local\Temp\14552005.tmp
 Input  Channels:   1   2   3   4   5   6   7
 Output Channels:   8   9  10  11  12  13  14
 Eigenchannels  :   1   2   3   4   5   6   7

 Sample Size      :          8867988

  Channel                Mean        Deviation

      1             9633.38         332.882
      2             6144.09         1575.29
      3             7410.84         3026.55
      4             10599.6         6349.97
      5             6561.08         796.338
      6             7574.04         750.672
      7             8585.73         409.603

 Eigenchannel     Eigenvalue       Deviation       %%Variance

      1            5.024e+07           7088            94%
      2            2.895e+06           1702           5.4%
      3            2.552e+05          505.2          0.48%
      4            2.282e+04          151.1         0.043%
      5            1.685e+04          129.8         0.032%
      6                 8560          92.52         0.016%
      7                820.8          28.65        0.0015%

Scaling Information:

 Eigen Output        ----Unscaled----     Deviation    Midpoint    Scale
 Channl Chnnl        Min          Max        Range                Factor

     1      8        -7083   8.612e+04 all     3.28e+04            1
     2      9        -5782     7.7e+04 all     3.28e+04            1
     3     10   -1.962e+04   5.792e+04 all     3.28e+04            1
     4     11        -6490    1.29e+04 all     3.28e+04            1
     5     12   -1.065e+04   1.106e+04 all     3.28e+04            1
     6     13    -1.02e+04        6079 all     3.28e+04            1
     7     14        -3175        5734 all     3.28e+04            1
```

***Figure 7.*** The PCA output log for the PCA RGB composite. The red outlined values are the variance percentages for each eigen channel band. The majority of the information (variance between 7 bands) is contained within the first eigen band, at 94%. The most contrast between cover types is found in the first band and the bands become noisier as the band numbers increase. This PCA output log was created with CATALYST Professional software.

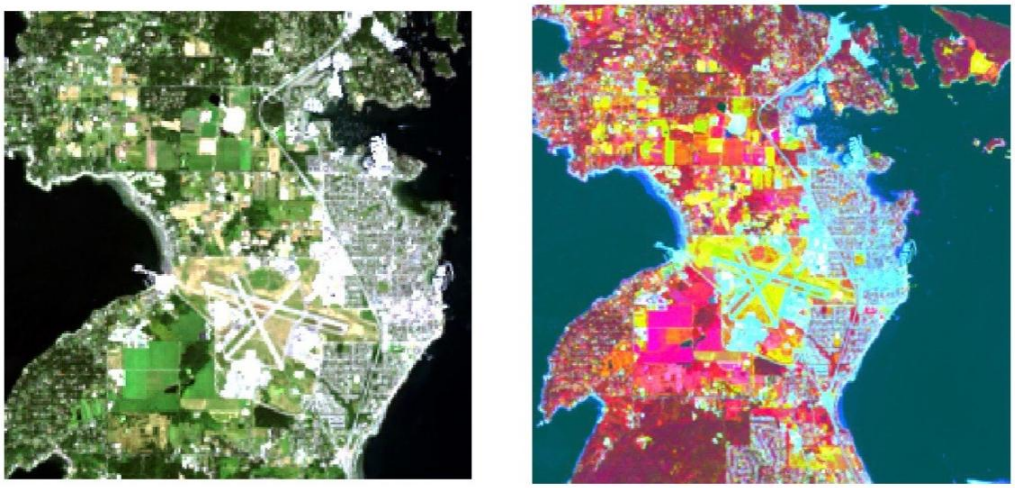


True Colour RGB PCA RGB

***Figure 8.*** A true colour vs PCA RGB composite. The advantage of the PCA RGB composite is that various land cover types may be easier to distinguish, compared to the true colour image. In this figure, the lake on the right appears green, as if it is land, but in the PCA image, it is the same blue as the water in the ocean. These composites were created with CATALYST Professional software.

True Colour RGB PCA RGB

***Figure 9.*** A true colour vs PCA RGB composite. The advantage of the PCA RGB composite is that various land cover types may be easier to distinguish, compared to the true colour image. There are multiple shades of green landcover in the true colour image but it is difficult to identify what each green represents. In the PCA image, the colours are clearly different. These composites were created with CATALYST Professional software.



True Colour RGB PCA RGB

***Figure 10.*** A true colour vs PCA RGB composite. The advantage of the PCA RGB composite is that various land cover types may be easier to distinguish, compared to the true colour image. In the true color image, some field appear the same white colour as the urban areas but in the PCA image, the fields are yellow and pink, while the urban areas are prominently cyan. These composites were created with CATALYST Professional software.

There is 94% of the original LANDSAT 8 data variance in eigen channel 1, as shown in *figure 7*, above.

The advantage of the PCA RGB composite, over the true colour image, is the ability to discern landcover types with more certainty. True color images can have different landcovers appear the same or nearly the same hue, but in a PCA RGB image, they are clearly different. Landcover types that are easy to identity in PCA include, fields or vegetation of nearly the same shade of green, urban and high albedo areas, lakes, and lakes. In *figure 8*, the lake on the right appears green and may be difficult to discern whether it is a body of water or a field. With the PCA image, it is apparent that the lake is indeed a lake, based on the bright blue hue. There are multiple shades of green landcover in *figure 9* but they may be difficult to identify what each colour represents. In the PCA image, the colours are clearly different. Different types of plans may reflect light differently, and the PCA image can show that better than the true colour image. *Figure 10* shows some fields appear bright white, as do the urban areas. The fields in the PCA image yellow and pink, while the urban areas are cyan. These figures evidently demonstrate the benefits of using a PCA RGB composite to determine landcover types in remote sensing.

References

Evans, T. (2023). *Geog 228 – Lab 5 Enhancements & Transformations*. Department of Geography, University of Victoria.

Osbey, M. (2025, February 6). *SPECTRAL VEGETATION INDICES (SVIs)*. ccpo.odu.edu. http://www.ccpo.odu.edu/SEES/veget/class/Chap_4/4_5.htm

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DEPARTMENT OF GEOGRAPHY, UNIVERSITY OF VICTORIA

LAB CREATION: TERRI EVANS

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